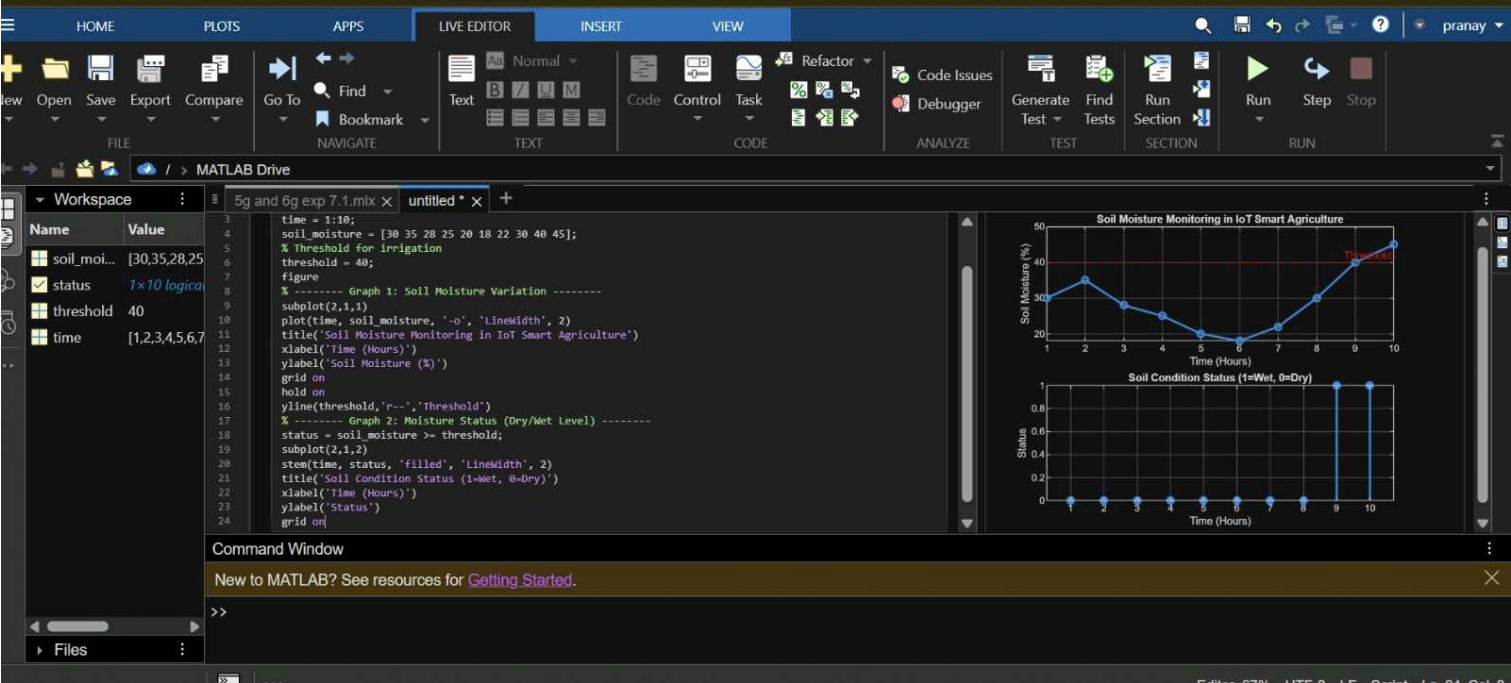
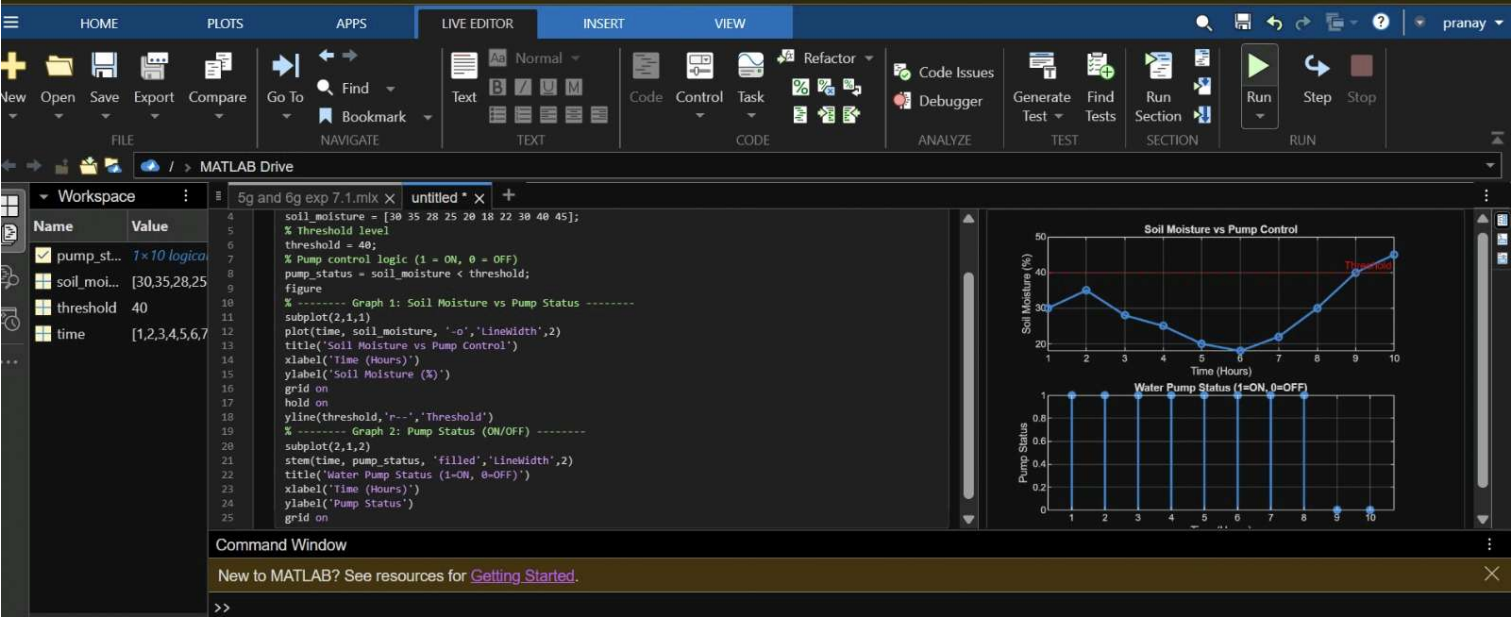






HOMEPLOTSAPPSLIVE EDITORINSERTVIEW

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FILE

NAVIGATE

TEXT

CODE

ANALYZE

TEST

SECTION

RUN

pranay

Workspace

Name	Value
efficiency	[75,87,5000,
soil_moi...	[30,35,28,25
threshold	40
time	[1,2,3,4,5,6,7

5g and 6g exp 7.1.mlx x

untitled * x

```
5 % Threshold level
6 threshold = 40;
7 % Irrigation efficiency calculation (%)
8 efficiency = (soil_moisture ./ threshold) * 100;
9 efficiency(efficiency > 100) = 100; % limit to 100%
10 figure
11 % ----- Graph 1: Soil Moisture vs Efficiency -----
12 subplot(2,1,1)
13 plot(time, soil_moisture, '-o','LineWidth',2)
14 title('Soil Moisture Variation')
15 xlabel('Time (Hours)')
16 ylabel('Soil Moisture (%)')
17 grid on
18 hold on
19 yline(threshold,'r--','Threshold')
20 % ----- Graph 2: Irrigation Efficiency -----
21 subplot(2,1,2)
22 plot(time, efficiency, '-s','LineWidth',2)
23 title('Irrigation Efficiency (%) in Smart Agriculture')
24 xlabel('Time (Hours)')
25 ylabel('Efficiency (%)')
26 grid on
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

>>

Files

Soil Moisture Variation

Time (Hours)	Soil Moisture (%)
1	30
2	35
3	28
4	25
5	20
6	18
7	25
8	30
9	35
10	40

Irrigation Efficiency (%) in Smart Agriculture

Time (Hours)	Efficiency (%)
1	75
2	87
3	50
4	40
5	30
6	25
7	50
8	75
9	87
10	100